Jashore University of Science and Technology

Jashore-7408, Bangladesh

Report on Internship
at Bangladesh Research and Education Network (BdREN)

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This is to certify that **Ajmain Muhtady Joarder (**ID: **190127)**, a student of the Department of Computer Science and Engineering, Jashore University of Science and Technology, has completed an internship at Bangladesh Research and Education Network (BdREN) from *4 February 2025* to *15 May 2025*. During this period, the intern was assigned to Network Operations Centre and BdREN Office, and was involved in various practical tasks. We found him to be sincere, punctual, and eager to learn.

Signature of the Student

Signature of the Supervisor

Shamim Ahmed
Network Engineer (NOC)
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Chapter One

Introduction

Chapter One: Introduction

In the rapidly evolving field of Information and Communication Technology (ICT), theoretical knowledge must be reinforced with practical exposure to produce industry-ready professionals. Recognizing this, academic institutions globally, including in Bangladesh, are increasingly encouraging students to undergo internships in real-world settings. Internships are designed not only to apply classroom concepts to practical scenarios but also to cultivate critical thinking, professionalism, and technical confidence. For a final-year Computer Science and Engineering student, an internship provides a crucial bridge between academic training and industry expectations.

This report represents a comprehensive reflection on my professional journey during an intensive internship at the Bangladesh Research and Education Network (BdREN). Established under the University Grants Commission of Bangladesh and funded by the World Bank, BdREN operates as the national research and education network (NREN), playing a pivotal role in Bangladesh's academic and scientific development.

Over the course of my internship, I had the privilege of working in two principal domains: the Network Operations Centre (NOC), housed in the UGC building, and the configuration and training unit at the central BdREN office. These locations are at the heart of BdREN's service delivery model, supporting more than 100 public and private institutions nationwide. From managing sophisticated data center systems to configuring network devices and running network simulations, my experience was both diverse and deeply enriching.

The primary aim of this report is to document and critically analyze my internship journey, elaborate on the tasks performed, tools used, and skills acquired, and provide valuable insights that may guide future interns, faculty, and network professionals alike. It also aims to highlight the broader significance of BdREN's contribution to academic digital infrastructure and suggest ways in which its internship program can be enhanced for greater impact.

1.1 Organizational Overview: BdREN

1.1.1 Historical Background and Formation

The Bangladesh Research and Education Network (BdREN) was established to serve a pressing need: to unify the ICT infrastructure of the country's higher education and research institutions. With rapid advancements in internet technologies, the global academic community increasingly relies on robust network backbones to share knowledge, conduct collaborative research, and host large-scale digital resources. Until the late 2000s, Bangladeshi universities operated in digital silos, with limited bandwidth and few mechanisms for inter-institutional communication.

To address this, the UGC initiated the BdREN project in 2009 with financial backing from the World Bank. The project was inspired by global best practices from entities like Internet2 (USA), GEANT (Europe), and TEIN (Asia-Pacific). BdREN was initially a government program but evolved into a trust to enable greater flexibility, innovation, and sustainability. As of 2025, BdREN connects over 100 academic institutions and continues to expand its infrastructure with increasing demands for remote learning, high-performance computing, and global collaboration.

1.1.2 Organizational Vision and Strategic Goals

BdREN envisions a future where every educational and research institution in Bangladesh is interconnected via a secure, reliable, and high-speed digital network. Its mission extends beyond basic connectivity, aiming to empower institutions with a full spectrum of ICT services that cater to the demands of modern academia. Key strategic goals include:

- Supporting national research through data accessibility and computation resources
- Enhancing the ICT readiness of universities through targeted training
- Bridging the digital divide among institutions by subsidizing high-speed access

- Encouraging international collaboration through partnerships with TEIN, GEANT, and APAN

BdREN's governance structure includes a Board of Trustees, composed of prominent academicians, government officials, and ICT leaders. This ensures that BdREN's policies and directions remain aligned with both national priorities and global trends in education and research.

1.1.3 Core Services and Offerings

BdREN's suite of services is designed to transform how institutions manage their digital ecosystems. These services can be categorized into infrastructure-level, application-level, and capacity-building services.

1. Infrastructure-Level Services

- **High-Speed Internet and Bandwidth Provisioning:** BdREN acts as a national ISP for educational institutions, offering rates far more competitive than commercial providers.
- **Layer 2 and Layer 3 VPNs:** Enable secure site-to-site connections and VLAN configurations across campuses.
- **Redundant Network Backbone:** Fiber-optic rings with built-in failovers for uninterrupted service.

2. Application-Level Services

- **Cloud Hosting and IaaS:** Institutions can host their websites, e-learning platforms, and applications on BdREN's cloud.
- **Zoom & Video Conferencing Integration:** Especially vital during the COVID-19 pandemic, BdREN facilitated online education for over 80 institutions.

- **IP Telephony & Unified Communications:** Reduces operational costs by transitioning from analog to digital communications.

3. Capacity Building and Technical Support

- **Training Workshops:** Covering cybersecurity, server management, and network optimization.
- **Consultancy Services:** For campus network planning and disaster recovery modeling.
- **Technical Documentation and Remote Support:** For real-time issue resolution and network diagnostics.

1.1.4 Geographic and Physical Presence

BdREN's headquarters is based in Dhaka, but its operational footprint spans the entire nation. With over 100 Points of Presence (PoPs) located in universities, colleges, and research centers, BdREN ensures that even institutions in rural or underdeveloped regions can access global academic resources.

The two key facilities that support BdREN's service architecture are:

- **Primary Data Center (UGC Building):** This is the main site responsible for traffic routing, cloud hosting, and network monitoring.
- **Disaster Recovery Center (BUET):** Mirrors the main site to provide redundancy and ensure operational continuity during outages.

1.1.5 International Linkages and Collaborations

One of BdREN's proudest achievements is its role in placing Bangladesh on the global map of academic networking. As a member of international networks like:

- APAN (Asia-Pacific Advanced Network)

- TEIN (Trans-Eurasia Information Network)
- GEANT (Europe's leading NREN)
- Internet2 (USA)

BdREN facilitates international research projects, cross-border data exchange, and access to repositories such as JSTOR, SpringerLink, and open-access journals.

Chapter Two

Internship Structure and Objectives

Chapter Two: Internship Structure and Objectives

2.1 Overview of the Internship Program

The internship at BdREN is structured to offer final-year undergraduate students a holistic introduction to professional network management, systems design, and real-time data center operations. The program spans multiple weeks and is carefully segmented into practical modules conducted across different facilities. Interns rotate through hands-on labs, configuration tasks, operational simulations, and observational roles, ensuring a rounded experience that combines conceptual clarity with applied expertise.

Unlike standard internships that often restrict students to observation, BdREN's program emphasizes engaged participation, encouraging interns to collaborate with engineers, operate testing environments, and analyze live infrastructure under guidance. The mentorship framework includes briefings, debriefings, technical documentation reviews, and daily assignments.

2.2 Specific Internship Modules

The internship was divided into two key modules, each designed with its own focus area and learning outcomes.

Module 1: Network Operations Centre (NOC)

Location: UGC Building, Dhaka

Duration: Approximately 1 month

Objective: Understand national-scale data center infrastructure, network reliability mechanisms, environmental control systems, and issue management workflows.

Key Learning Tools: SolarWinds NPM, UPS & PAC systems, raised floor layouts, manual SLA documentation, mock ticketing systems.

Learning Activities:

- Monitoring simulated and live alerts on performance dashboards.

- Reviewing architecture of BdREN's national topology.
- Observing generator and HVAC failover scenarios.
- Conducting role-based drills to simulate critical outages.

Module 2: BdREN Head Office

Location: Navana HR Tower-2, Level-5, 205/1 Bir Uttam Mir Shawkat Sarak, Tejgaon-Gulshan Link Road, Dhaka 1208

Duration: Approximately 2 months

Objective: Acquire hands-on configuration skills for enterprise-grade routers and switches, simulate complex topologies, and explore protocol behavior in controlled settings.

Key Tools: Huawei AR2200, Cisco Packet Tracer, manual CLI configuration, subnet calculators, routing tables.

Learning Activities:

- Performing router resets and configuring from scratch.
- Building static and multi-router simulated networks.
- Conducting cross-subnet communication tests.
- Debugging packet loss and interface misconfigurations.

2.3 Work Ethic

Throughout the internship, maintaining a strong work ethic was essential not only for individual growth but also for contributing meaningfully to team-based activities. The structured and professional environment at BdREN demanded consistency, discipline, and accountability—traits that I actively developed and demonstrated during the internship.

- **Punctuality and Commitment:** I consistently adhered to the assigned schedules and arrived on time for daily briefings, simulations, and lab sessions. This routine helped build a disciplined mindset and demonstrated reliability to both mentors and teammates.

- **Accountability:** Whether I was responsible for configuring a router interface, documenting a simulation scenario, or presenting group findings, I took ownership of every task. I ensured that my individual contributions were accurate, well-prepared, and aligned with team objectives.

- **Attention to Detail:** Network configuration tasks often require exact parameter settings. I developed a habit of double-checking IP addresses, subnet masks, and routing commands to minimize errors. This meticulous approach helped prevent avoidable issues during practical sessions.

- **Self-Initiative:** In moments where direct guidance was unavailable, I proactively researched solutions, experimented with settings, and consulted official documentation or CLI help menus. This initiative not only expanded my learning but also reduced unnecessary dependency on others.

- **Consistency in Learning:** I made it a point to record daily learnings and review them regularly. This consistent effort improved retention and allowed me to build on previously acquired knowledge as the internship progressed.

- **Respect for Team Roles:** During group-based tasks, I respected each member's assigned role and focused on mutual support rather than competition. This collaborative mindset helped build a positive work culture within the intern cohort.

This commitment to professional work habits helped me adapt to BdREN's structured environment and reinforced the discipline required to operate within real-world ICT infrastructures.

Chapter Three

My Involvement in Office Work

3.1 Internship Goals and Deliverables

Each intern was required to:

- Maintain a technical logbook documenting daily learning outcomes.
- Submit network diagrams and configuration snapshots.
- Collaborate in small teams for simulation projects.
- Produce a final internship report summarizing technical insights and growth reflections.

This structured approach ensured we not only learned how but also understood why — a critical mindset for sustainable technical growth.

3.2 Network Operation Centre (NOC) – UGC Building

The NOC is BdREN's most critical operational site. It is the heartbeat of real-time network management, hosting equipment and systems that support nearly every academic institution in Bangladesh.

3.2.1 Infrastructure Overview

The data center operates 24/7, monitored by a team of engineers and administrators. It is divided into various zones, including:

- **Server Zone:** Houses the core computational and application servers.
- **Switch Zone:** High-performance managed switches route internal and external traffic.
- **Power Zone:** Equipped with generators, rectifiers, and UPS backup systems.
- **Cooling Zone:** Uses PAC systems for environmental control.

- **Monitoring Bay:** Features dashboard walls and terminals for real-time alerts.

We observed various rack-mounted devices, including:

- Redundant authentication servers
- Proxy servers
- DNS servers with failover routing
- NTP (Network Time Protocol) servers
- Firewalls enforcing regional access policies

3.2.2 HVAC, Power and Raised Flooring Systems

One of the key learning takeaways was understanding the importance of power redundancy and climate control in large data centers. We explored:

- **Dual Generator Setup:** G1 (300kVA) and G2 (100kVA) connected via ATS.
- **Sub Distribution Boards (SDB):** Balanced power load across racks.
- **Uninterruptible Power Supply (UPS):** Ensures seamless transition to backup power for approximately one hour.
- **Rectifiers:** Convert AC to DC for stability.
- **PAC Systems:** Precision cooling units maintaining a narrow temperature/humidity band.

The raised floor system serves dual purposes:

1. **Airflow Routing:** Cooled air is channeled under floor tiles and released through perforated panels.
2. **Cable Management:** Reduces clutter and enhances safety by isolating power and data cables.

3.2.3 Network Architecture and Data Flow

BdREN follows a tiered network architecture, allowing efficient and scalable data handling:

1. **Upstream Router (Tier 1):** Links BdREN to international networks like GEANT or APAN.
2. **Core Router (Tier 2):** Central node handling the main routing logic.
3. **Distribution Router (Tier 3):** Routes traffic to regional or city-level clusters.
4. **Edge Router (Tier 4):** Interfaces directly with institutional local networks.

All routers are interconnected via high-throughput optical fiber, using:

- Single-mode fiber for long-distance transmission
- Dense Wavelength Division Multiplexing (DWDM) for bandwidth efficiency

3.2.4 Ticketing System and Network Monitoring

Monitoring tools allow NOC engineers to:

- Track real-time bandwidth usage
- Detect packet drops, jitter, and latency spikes
- Identify downed interfaces or high CPU loads

We engaged with the SolarWinds NPM software to:

- Configure custom alerts for simulated outages
- Learn SNMP polling basics and MIB object interpretation
- Analyze traffic spikes during mock DDoS attacks

The ticketing system assigns SLA-bound tasks to engineers. Each ticket tracks:

- Type (connectivity, performance, DNS failure, etc.)
- Severity level
- Assigned engineer
- Resolution time and report

We simulated two full ticket cycles: one for fiber disconnection, another for service timeout at a PoP site.

3.3 BdREN Head Office

This phase of the internship provided hands-on experience with real-world network device configurations and protocol implementations. We also gained exposure to working in a professional office environment and developed essential communication and collaboration skills. Some of the configurations we worked on are shared below.

Router Configuration: Huawei AR2200

3.3.1 Introduction to Huawei Routers

Huawei's AR2200 series is a versatile family of enterprise routers used in many educational institutions due to their cost efficiency and strong performance across Layer 3 tasks. These routers support static routing, dynamic routing protocols (OSPF, BGP), VPNs, and QoS, making them ideal for lab-based and real-world applications.

During the internship, each team was provided with a Huawei AR2200 unit, a USB-to-console cable, and a laptop pre-installed with PuTTY, the go-to terminal emulator for network engineers.

3.3.2 Connection and Console Setup

Procedure:

1. **Hardware Setup:**
   - The USB end of the console cable was inserted into the laptop.
   - The RJ45 end was plugged into the console port of the router.
2. **Driver Installation:**
   - Ensured USB-to-Serial drivers were recognized by the OS.
   - Used Windows Device Manager to identify the COM port (e.g., COM3).
3. **Launching PuTTY:**
   - Set connection type: Serial
   - Input COM port (e.g., COM3), speed: 9600
   - Opened session to initiate communication

3.3.3 Factory Reset and CLI Configuration

We simulated a password reset and factory configuration:

- Entered boot menu during startup
- Selected option 6: "Reset Password"
- Chose option 0 to return to default menu
- Rebooted router to activate reset

Initial login:

- Username: `admin`

- Password: `Admin@huawei`

Post-reset:

- Set new password: `@bDren1234`

Basic Configuration Steps:

```
system-view
sysname BdREN-Router
interface GigabitEthernet0/0/0
ip address 192.168.10.1 255.255.255.0
undo shutdown
quit
ip route-static 192.168.20.0 255.255.255.0 192.168.10.2
save
```

These steps configured the interface and set up a static route. The `undo shutdown` command activated the interface.

3.4 Cisco Packet Tracer Labs

Cisco Packet Tracer was central to the simulation modules. It allowed us to build complex topologies, visualize traffic flow, and practice failure recovery without impacting physical hardware.

We completed two major labs:

3.4.1 5.4 Static Routing with Single Router

Topology:

- One Router (Router0)

- Two Switches (Switch0 and Switch1)
- Four PCs split across two subnets

Addressing Table:

Device	IP Address	Subnet Mask	Default Gateway
PC0	192.168.0.1	255.255.255.0	192.168.0.3
PC1	192.168.0.2	255.255.255.0	192.168.0.3
PC2	172.16.0.1	255.255.255.0	172.16.0.3
PC3	172.16.0.2	255.255.255.0	172.16.0.3

Router Configuration:

```
interface FastEthernet0/0
ip address 192.168.0.3 255.255.255.0
no shutdown

interface FastEthernet0/1
ip address 172.16.0.3 255.255.255.0
no shutdown

ip route 172.16.0.0 255.255.255.0 FastEthernet0/1
ip route 192.168.0.0 255.255.255.0 FastEthernet0/0
```

Verification:

- Pinged PC0 to PC2: successful round-trip
- Validated routing with `show ip route`

3.4.2 5.5 Static Routing with Dual Routers

Topology Highlights:

- Two routers interconnected via 10.0.0.0/30 subnet
- Two VLANs: 192.168.1.0/24 and 192.168.2.0/24
- Each router handled a different subnet

Router IPs:

- Router0: 192.168.1.1 (LAN), 10.0.0.1 (WAN)
- Router1: 192.168.2.1 (LAN), 10.0.0.2 (WAN)

Routing Commands:

Router0:

```
ip route 192.168.2.0 255.255.255.0 10.0.0.2
```

Router1:

```
ip route 192.168.1.0 255.255.255.0 10.0.0.1
```

Observations:

- Tested failover by simulating a cable disconnect
- Noted TTL changes during multi-hop pings
- Measured latency via ICMP timing

3.4.3 Additional Activities

- Drafted a sample network migration plan for a fictional university
- Created an addressing table for a 50-node network using VLSM
- Discussed NAT (Network Address Translation) and its applications in home routers and institutional gateways

3.5 Tools and Technologies Used

During the internship, we worked with a range of tools and technologies that are fundamental to real-world network management and configuration.

3.5.1 PuTTY (Terminal Emulator)

PuTTY is an open-source terminal emulator widely used by network administrators to establish console and SSH sessions with network devices. In our internship:

- We used PuTTY to access Huawei AR2200 routers through USB-to-Console connections.
- It allowed us to configure devices without needing a web interface or GUI.
- We learned how to adjust baud rate settings, serial line configurations, and session logging.

3.5.2 SolarWinds Network Performance Monitor (NPM)

This enterprise-grade monitoring platform enabled us to:

- Monitor live network performance: Interface traffic, device uptime, and bandwidth consumption.
- Configure alerts: Thresholds for packet loss, CPU temperature, or uptime could trigger emails or SMS alerts.
- Understand SNMP polling: We studied the polling intervals and learned how to interpret SNMP MIBs (Management Information Bases).

We also explored how SolarWinds integrates with Cisco and Huawei routers using SNMPv2/v3, offering a unified view of both hardware and link health.

3.5.3 Cisco Packet Tracer

This network simulation software was the most interactive tool we used:

- Created and saved multiple topologies from scratch.
- Simulated routing protocols like static routing and RIP.
- Observed data flow via animation, helping us understand packet paths.
- Diagnosed problems using simulation mode before transferring concepts to physical labs.

3.5.4 Huawei AR2200 Router

Our practical training centered around the AR2200, which is equipped with:

- Multiple Gigabit Ethernet ports
- USB and RJ45 console access
- Support for firewall policies and ACLs
- Rich routing protocol support including OSPF and BGP

Chapter Four

Challenges and Solutions

4.1 Challenges Faced and Solutions

The transition from classroom simulation to configuring real devices and monitoring live infrastructure was not without its challenges. Here, I document the key technical and procedural issues I faced, alongside the strategies used to resolve them.

4.1.1 Hardware Interface Errors

Problem: While connecting the console cable, PuTTY failed to detect the correct COM port.

Solution:

- Installed updated USB-to-Serial drivers.
- Used Device Manager to verify the COM number.
- Restarted PuTTY session with correct port settings (e.g., COM3, 9600 baud rate).

4.1.2 Router Reset Fails

Problem: After factory reset, the router booted into default mode without enabling CLI configuration.

Solution:

- Power-cycled the router.
- Re-entered the boot menu.
- Re-selected option 6 and saved settings after the password change.

4.1.3 VLAN Misconfigurations

Problem: PCs on the same switch were unable to communicate despite having the same subnet.

Solution:

- Identified that the access ports were not assigned to the correct VLAN.
- Applied port assignment using:

```
switchport mode access
switchport access vlan 10
```

4.1.4 Static Routing Failures

Problem: Inter-router communication failed in the Day 2 simulation.

Solution:

- Used `show ip route` to debug entries.
- Found incorrect subnet mask (255.255.255.252 instead of 255.255.255.0).
- Re-entered routes with correct masks and verified via ICMP.

4.1.5 7.5 High Latency in Simulation

Problem: Packet delay was too high in Packet Tracer tests.

Solution:

- Reduced simulation hops.
- Disabled unnecessary protocols.
- Split tasks across VLANs to reduce traffic load.

Chapter Five

Professional Development

Chapter Five: Professional Development

5.1 Skills Gained and Learning Outcomes

This internship has significantly enriched both my technical and interpersonal skillset. The exposure to enterprise-scale networking in a national environment was deeply impactful. Key skills developed include:

- Hands-on configuration of routers and switches
- Network simulation and logical topology design
- Troubleshooting routing and VLAN issues
- Using monitoring tools for real-time diagnostics
- Collaborating effectively within technical teams
- Preparing documentation, diagrams, and reporting outputs

5.2 Technical Skills Acquired

5.2.1 Device Configuration

- Performed factory resets, initial configuration, and static routing setup on physical routers.
- Implemented interface-based addressing and port enabling on switches.

5.2.2 Protocol Mastery

- Developed strong foundational knowledge of static routing, subnetting, and VLAN segmentation.
- Understood how ARP requests, MAC tables, and STP work in real-time topologies.

5.2.3 Simulation and Troubleshooting

- Used simulation tools to map complex topologies before hardware deployment.
- Identified broken routes, misconfigured interfaces, and missing gateways.

5.2.4 Monitoring and Alert Management

- Set up alert systems in SolarWinds.
- Interpreted SNMP responses and analyzed MIB metrics.

5.3 Professional and Soft Skills

In addition to technical development, the internship at BdREN significantly enhanced a variety of professional and soft skills essential for success in the networking and ICT field:

- **Communication:** Regular technical discussions, daily briefings, and simulation walkthroughs helped sharpen my ability to convey complex networking concepts clearly. I learned to explain configuration logic, describe problems during simulations, and propose solutions in a structured and articulate manner. These practices strengthened both my verbal and written communication within technical environments.
- **Documentation:** Maintaining a structured technical logbook throughout the internship allowed me to document daily learning outcomes, device configurations, and troubleshooting steps. This habit improved my precision and taught me how to maintain consistent records—an essential skill in any production-grade network environment where every change must be traceable and well-documented.
- **Team Collaboration:** Many tasks during the internship, including Packet Tracer simulations, VLAN configuration, and failover drills, were performed in teams. I gained experience dividing responsibilities, validating peer configurations, and working collaboratively

under time constraints. These activities closely mirrored real-world project environments, where coordination and teamwork are vital.

- **Critical Thinking and Problem Solving:** The internship placed us in situations where predefined instructions were unavailable, requiring us to think independently. When facing misconfigurations, interface errors, or routing failures, I learned to analyze CLI outputs, verify topology logic, and apply structured problem-solving techniques. This helped me develop confidence in addressing unexpected issues without relying on direct supervision.

- **Professional Etiquette and Work Culture:** Working inside a nationally significant ICT organization introduced me to workplace etiquette such as punctuality, respectful communication, and adherence to technical protocols. Whether interacting with senior engineers, responding to ticketing system alerts, or attending scheduled review meetings, I experienced how formal behavior and discipline contribute to the stability of large-scale network operations.

Chapter Six

Self Assessment

Chapter Six: Self Assessment

Reflecting on my internship experience at BdREN, I believe I demonstrated consistent dedication, adaptability, and a proactive approach to learning. I entered the program with foundational knowledge in networking, but this internship allowed me to apply those concepts in real-world environments and significantly broaden both my technical and professional capabilities.

One of my key strengths during the internship was my willingness to take initiative. Whether configuring devices, troubleshooting network issues, or exploring monitoring tools like SolarWinds, I made a conscious effort to go beyond basic instructions. I frequently reviewed command-line outputs, explored documentation, and sought out additional resources to better understand the underlying technologies. This self-driven learning helped me become more confident in identifying and resolving practical network problems.

I also maintained strong consistency in task completion and documentation. By recording configurations, logging daily learnings, and reviewing each session's outcomes, I ensured that my progress was structured and traceable. This habit not only improved my retention but also allowed me to contribute meaningfully during team discussions and simulations.

While I was comfortable with static routing, basic VLAN setup, and simulation tools, I recognized a gap in my familiarity with more advanced protocols such as OSPF, BGP, and ACL configuration. Though these topics were touched upon, I believe further exposure and independent practice are needed to gain full confidence in deploying such protocols in production-grade environments.

From a professional standpoint, I adapted well to the formal culture at BdREN. I was punctual, respectful in communication, and cooperative in group settings. I actively participated in discussions, accepted feedback positively, and supported team members whenever possible.

In summary, I believe I made the most of this internship opportunity. I grew not only in terms of technical proficiency but also in my ability to work in a structured, team-based environment. The experience has inspired me to pursue deeper learning in network security, protocol design, and cloud infrastructure, with the goal of becoming a well-rounded network professional.

Observations

Throughout the internship, several key observations stood out. First, the structured rotation between the NOC and the head office provided a balanced exposure to both monitoring operations and device-level configurations. The hands-on nature of the tasks allowed for a much deeper understanding of theoretical concepts learned in university. For example, configuring static routes on physical routers reinforced the logic behind subnetting and routing decisions.

Second, the professionalism and teamwork within BdREN created a productive learning environment. Engineers were approachable and willing to guide us through complex issues, which significantly improved our troubleshooting confidence. It was also clear that documentation and consistent communication play a vital role in managing large-scale network operations.

Finally, while the internship covered a wide range of topics, more time on advanced routing protocols or security practices could enhance the learning experience. Overall, the program was well-structured, informative, and highly relevant to real-world networking scenarios.

Chapter Seven

Conclusion

Chapter Seven: Conclusion

This internship marked a significant milestone in my journey toward becoming a proficient network engineer. At BdREN, I wasn't just a passive observer but an active contributor to a national ICT framework that empowers learning and innovation across Bangladesh.

The two-phase structure combining NOC insight and head office learning environment offered a comprehensive view of both backend infrastructure and hands-on device management. I gained practical skills in router configuration, switch segmentation, VLAN deployment, and network performance monitoring. I also improved my ability to communicate technically, work under pressure, and resolve problems systematically.

In conclusion, I believe this experience has strengthened my readiness to contribute to professional network environments. It has inspired me to pursue further studies and certifications in network security and cloud networking. I am grateful to BdREN, my supervisors, and my peers for making this internship not only educational but truly transformative.